Martin Wong

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Education

Simon Fraser University Expected Graduation April 2027
Bachelor of Science, Computer Science *Burnaby, BC*
Courses: Data Structures and Algorithms, Database Systems, Software Engineering, Distributed Systems, Computer Networking

Work Experience

Safety CLI Cybersecurity May 2026 – August 2026
Incoming Software Engineer Intern *Vancouver, BC*

Electronic Arts May 2025 – August 2025
Software Engineer Intern *Burnaby, BC*

- Engineered and deployed an intelligent internal tooling system with **LLM RAG pipeline** using **Python**, cutting onboarding time and helping 30+ engineers quickly resolve codebase and documentation questions
- Built scalable vector embedding pipeline indexing millions of lines of production code and 300+ documentation pages using token-based chunking with **AWS Bedrock** embeddings stored in **PostgreSQL** with **pgvector**
- Designed a custom **MCP server** with a scoring system to dynamically rank and retrieve from docs, code, chat history, and user feedback, boosting positive user feedback from **29%** to **85%**
- Reduced infrastructure costs by 35% by deploying **FastAPI** on **AWS EKS** with **pgvector** on **Aurora**, leveraging on-demand **AWS Bedrock** inference and optimized connection pooling over managed vector search services

Rivian & Volkswagen Group Technologies September 2024 – April 2025
Software Engineer Intern *Vancouver, BC*

- Led mobile development of a 2FA-secured driving feature on **iOS** and Android using **Swift** and **Kotlin**, enabling secure remote commands across all Rivian vehicles directly affecting **10,000+ customers** 32
- Implemented fault-tolerant command delivery via **APNS** and **Firebase Messaging** with Bluetooth byte/Protobuf fallback, and built time-based one-time passcodes using **HMAC encryption** and **GraphQL** for all notification failure scenarios
- Built native **iOS screen recording tooling** using **AVFoundation** and **Core Graphics** for automated telemetry capture, deployed via **App Center** to 900+ test devices for issue reproduction and debugging

Avena Tech Corporation July 2023 – September 2024
Software Developer (Part-Time) *Vancouver, BC*

- Streamlined deployment by utilizing Bash, Github Actions, and **Docker** to create a **CI/CD** pipeline for testing, building, and deploying containers, reducing development time by 50%
- Architected a microservices-based platform with containerized REST API endpoints using **Express.js**, **AWS S3**, and PostgreSQL, migrating 60+ WordPress blogs to a new efficient system
- Restructured VPS infrastructure and provisioned cloud servers with Docker runners and Docker Compose orchestration, reducing infrastructure costs by 40% while enabling custom GitHub Actions runners for accelerated CI workflows

Simon Fraser University iXLab April 2023 – September 2023
Undergraduate Research Assistant *Vancouver, BC*

- Developed IMU sensor simulation system in **Python** for validating machine learning models, processing XYZ accelerometer and gyroscope data from T2M-GPT generated motion sequences

Projects

Encrypted P2P Chat App | *React, TypeScript, WebRTC, PeerJS, Go* |

- Architected a serverless P2P chat application with end-to-end **AES-256-GCM** encryption using **WebRTC**, implementing NAT traversal with STUN/TURN servers for 95%+ connectivity
- Designed distributed file transfer protocol with chunked streaming, automatic retry logic, and congestion control for reliable large file transfers over unreliable networks

Personalized AI Discord Bot | *Python, Node.js, FastAPI, AWS Bedrock* |

- Fine-tuned Meta's Llama2 LLM on **120,000** Discord messages to create a personalized chatbot to respond in the style of the user
- Trained and deployed the fine-tuned model on Replicate accessed by a FastAPI endpoint, containerized on Docker and hosted on an **AWS EC2 instance**, and deployed via PM2 on a personal home Ubuntu server for continuous uptime

Technical Skills

Languages: Java, Python, Go, Kotlin, JavaScript, TypeScript, C, SQL, Bash, Swift
Backend & Distributed Systems: Docker, Kubernetes, AWS, PostgreSQL, MongoDB, Redis, GraphQL, WebRTC
Frameworks/Libraries: React, Next.js, FastAPI, Express.js, Node.js, Jest
Developer Tools: Git, Postman, NGINX, Linux, CI/CD Pipelines, pgvector